U-T-00190A(DSA-DM)
9 May 1972
SUPERSEDING
Int. Fed. Spec. U-T-00190(DSA-DM)
1 December 1964

INTERIM FEDERAL SPECIFICATION

TEST PAPER AND COLOR CHART, GLUCOSE IN URINE, 100 Tests

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa. 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 This specification covers test paper sufficient for 100 tests, in a roll-type dispenser with an affixed color chart, for use in the qualitative and quantitative determination of the presence of glucose in urine. Potency: 24 Months. (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566	Box, Folding, Paperboard.
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-Plywood.
PPP-B-621	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard
PPP-B-676	Boxes, Setup.
PPP-C-00186a	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

Federal Standard:

Fed. Std. 123	Marking for Domestic Shipment (Civilian Agencies).

FSC 6505

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(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547 Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, D. C. 20204.)

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3. REQUIREMENTS

3.1 Material.

3.1.1 Test paper. Shall be a roll of yellow paper tape, impregnated with purified glucose oxidase and peroxidase enzymes for detecting 0, 1/10, 1/4, 1/2, and 2 percent concentrations of glucose in urine, when tested as described in 4.5.1. The paper shall be approximately 1/4 inch in width and not less than 150 inches in length. When dispensed from the roll-type dispenser, the roll of paper tape shall be of sufficient overall length to permit the cutting or tearing off of not less than 100 pieces of test paper, each such piece being 1-1/2 inches in length. The test paper shall be marked to indicate when approximately four tests (6 inches of paper) remain in the dispenser. Shall be suitable for the qualitative and quantitative determination of glucose in urine.

3.1.2 Dispenser. Shall be a roll-type dispenser suitable for easy insertion into and withdrawal from the pocket or purse. Dispenser shall be approximately 2 inches square and approximately 1/2 inch in width and shall be equipped with an integral hinged lid which is capable of being snapped closed for protection of the tape, and which shall comply with lid (hinge) test described in 4.5.2. The dispenser shall be equipped with a suitable cutting edge or spike inside the lid to facilitate division of the tape into 1-1/2 inch lengths. The dispenser shall be of such design as to permit opening of the dispenser in the event the roll of test paper breaks and rethreading of the test paper through the dispensing outlet is necessary.

3.1.3 Color chart. Color chart shall be affixed to the dispenser (see 5.2.1). The color chart shall be calibrated for the lot or control of the test paper contained in the dispenser and shall show the colors which may develop when the urine being tested contains approximately 0% to 2% or more of glucose.

3.2 Delivery. Not less than 21 months of the expiration dating period (potency period) shall remain at the time of delivery of the product, to the Government.

3.3 Workmanship. Workmanship shall be first class throughout. The test paper and the dispensers shall be free from any defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, quantity of paper tape, produced from one formulation, subjected to the same manufacturing operation, and placed into the roll-type dispenser.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major A	S-3	1.0	Roll and Dispenser
Major B	S-3	1.0	Roll, Dispenser, and Unit Package

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4.3.1 For end item testing. A composite sample of S-3 shall be used. Where possible, the same sample shall be used for two (2) or more tests. Sufficient quantities shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

4.4 Examination. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major A

101.	Test paper not at least 150 inches in length.
102.	Test paper not of proper width for dispenser.
103.	Test paper not yellow in color.
104.	Test paper not marked to indicate the last 6 inches of paper remaining in the dispenser.
105.	Dispenser not proper dimensions.
106.	Dispenser not equipped with a cutting edge or spike to facilitate division of the test paper into 1-1/2 inch test lengths.
107.	Dispenser not fabricated to permit opening for rethreading of dispensers.
108.	Dispenser not roll-type.

Major B

151.	Labeling of immediate containers and unit package (enclosing carton) not legible.
152.	Labeling of immediate containers and unit package (enclosing carton) not complete.
153.	Circular or brochure not furnished as specified.

*Inspection is not restricted to the classified possible defects listed above.

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4.5 Tests.

4.5.1 Glucose sensitivity. Prepare standard solutions of 1/10, 1/4, 1/2, and 2.0 percent glucose in glucose-free urine. The solution shall be aged at least 30 minutes before testing. The four glucose standard solutions and the glucose-free (blank) solution shall each be tested as follows:

> Dip one end of a 1-1/2 inch piece of test paper in the standard solution. Remove and wait one minute for the color to develop. (For the 1/2 and 2.0 percent concentrations, wait two minutes).
>
> Compare the darkest area on the test paper with the color chart. The test papers, after dipping into each standard solution and blank, shall produce the same color as indicated on the color chart for the corresponding glucose concentration.

4.5.2 Lid (hinge) test. The lid of the dispenser shall remain affixed to the dispenser when the lid is fully opened and completely closed one hundred (100) times. At least five (5) dispensers from each lot shall be tested.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified herein and shall be in accordance with the applicable paragraphs of Interim Federal Specification PPP-C-00186a, dated 15 May 1969, and Amendment-1, dated 27 October 1969.

5.1.1 Immediate containers. Immediate containers (dispensers) shall be as specified in 3.1.2 of this document.

5.2 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug, and Cosmetic Act, and shall include the information required in 5.2.1.

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5.2.1 Immediate containers (dispensers). Each immediate container (dispenser) label shall bear the following information. However, the information is not required to appear in the sequence indicated.

(a) the commercial name

(b) the lot or control number

(c) the name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear.

When both names are placed on the label, the following designations shall precede the names:

"MFR" for the manufacturer and
"CONTR" for the contractor.

(d) a color chart calibrated for the lot or control of test paper contained in the dispenser showing the color which will develop when the urine being tested contains approximately 0%, 1/10%, 1/4%, 1/2%, and 2% or more of glucose.

The chart shall show a comparison of the color concentrations to the 0, +, ++, +++, ++++ readings commonly referred to in the utilization of Benedict Solution Test.

(e) the label which includes the color chart shall bear the following or similar statements:

1. Protect from direct light, excessive moisture, and heat.

2. DIRECTIONS ON BACK, in prominent red letters.

(See additional label information on page 8)

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(f) the label on the side opposite the one bearing the color chart shall include the following or similar information:

DIRECTIONS (in red print)

1. Lift top lid and withdraw approximately 1-1/2 inches of tape. While keeping a slight tension on tape, close lid and hold. Tear tape by pulling straight out.

2. Dip part of tape in specimen; remove and wait one minute. Yellow color indicates urine is sugar free.

3. Then immediately compare the darkest area with color chart on opposite side. If tape indicates 1/2% or higher, wait one additional minute and make final comparison.

(Words underlined shall be appropriately highlighted).

(g) the label on the side opposite the one bearing the color chart shall also include the following or similar information:

1. After opening carton, use tape within 4 months. Must be used prior to expiration date. (This shall be in red letters).

2. the expiration date designated by month, day, and year

3. the lot or control number.

Label adhesion test shall not apply.

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5.2.2 Unit packages. Each unit package label shall bear the following information. However, the information is not required to appear in the sequence indicated.

(a) the item identification designated as "TEST PAPER AND COLOR CHART, GLUCOSE IN URINE"

(b) the number of tests designated as "100 tests"

(c) the Federal Stock No.

(d) the lot or control number

(e) the name and address of the manufacturer. When the manufacturer is not the contractor, the name and address of the contractor shall also appear.

When both names are placed on the label, the following designations shall precede the names:

"MFR" for the manufacturer and
"CONTR" for the contractor.

(f) the expiration date designated by month, day, and year

(g) the following or similar statements:

1. Protect from direct light, excessive moisture, and heat (in red print).

2. Red mark on tape indicates that approximately four tests (6 inches) remain in the dispenser.

(h) directions similar to those which are required for the immediate container.

5.2.2.1 Circulars. A circular, brochure, or other printed matter shall be packaged within each unit package setting forth as a minimum: Instructions for use with appropriate pictures, and a suitable chart for the user to record the date, hour, and percent of glucose found when the tape was used.

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5.3 Packaging. Packaging shall be Level A or C, as specified. (See 6.1).

5.3.1 Unit of issue. One roll of test paper, as specified, constitutes one unit of issue.

5.3.2 Packaging quantities. The number of units of issue indicated in Table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection as specified in the procurement document.

Table I. Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	144 units

5.3.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.3.3 Level A.

5.3.3.1 Unit package. Each dispenser containing one roll shall be packaged in a carton of appropriate size and design constructed in accordance with PPP-B-566, except that commercial colors are acceptable. The carton shall be overwrapped in a suitable protective foil wrapper, and wrapper heat sealed.

5.3.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors are acceptable. Closure shall be adecuate to prevent accidental opening.

5.3.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.4 Packing. Packing shall be Level A, B, or C, as specified. (See 6.1).

5.4.1 Level A.

5.4.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.4.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.4.2 Level B.

5.4.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of Table I of PPP-B-636 shall apply for PPP-B-636 box.

5.4.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.5 Marking.

5.5.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.5.2 Military activities.

5.5.2.1 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Lot (control) number, contract or purchase order number, and name of contractor shall be shown. Type I Shelf-Life markings shall be shown as specified in MIL-STD-129. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)."

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5.5.2.2 Exterior containers. Exterior containers shall be marked in accordance with MIL-STD-129. Lot (control) number shall be shown. Type I Shelf-Life markings shall be shown as specified in MIL-STD-129. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)."

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification
(b) the Federal Stock No.
(c) applicable levels of packaging and packing (see 5.3 and 5.4).

6.2 This specification covers the following item appearing the Federal Supply Catalog:

FSN 6505-559-6859 TEST PAPER AND COLOR CHART, GLUCOSE IN URINE, 100 Tests

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1285

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

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NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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